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$$S_{2n}=\sum_{k=1,impair}^{2n}\frac{(-1)^{k-1}}{k}+\sum_{k=1,pair}^{2n}\frac{(-1)^{k-1}}{k}=\sum_{p=1}^{n}\frac{1}{2p-1}-\sum_{p=1}^{n}\frac{1}{2p}=\sum_{p=1}^{2n}(\frac{1}{2p-1}+\frac{1}{2p})-2\sum_{p=1}^{n}\frac{1}{2p}=\sum_{k=1}^{2n}\frac{1}{k}-\sum_{k=1}^{n}\frac{1}{k}=\sum_{k=n+1}^{2n}\frac{1}{k}\int_{n+1}^{2n+1}\frac{1}{t}d$$